Lecture 13

De Rham cohomology

Objectives. *A closed form of positive degree has local primitives, forms whose exterior derivative is the given form, but a global primitive may not exist. De Rham cohomology records this obstruction. We prove the local assertion by integrating along a contraction. We then compare homotopic maps and compute the cohomology of spheres by joining two open sets. The Coulomb form gives a nonzero class with a measurable period.*

By the end of this lecture you will be able to:

- Distinguish closed forms, exact forms and their cohomology classes.
- Construct a primitive of a closed positive-degree form on a star-shaped open set.
- Prove smooth homotopy invariance using the homotopy operator.
- Compute sphere cohomology from Mayer–Vietoris and normalize its area class.

Unit 13A Local primitives and homotopy

13.1 Closed forms modulo exact forms

All manifolds in this lecture are smooth and without boundary. All vector spaces and cohomology groups have real coefficients. Recall from Lecture 10 that a form α is *closed* if $\mathrm{d}\alpha = 0$ and *exact* if $\alpha = \mathrm{d}\eta$ for some form η. The identity $\mathrm{d}^2 = 0$ makes every exact form closed.

Definition 13.1. A *cochain complex* is a sequence of vector spaces with linear maps $\mathrm{d}\colon C^k \to C^{k+1}$ whose successive composites are zero.

The spaces $\Omega^k(M)$ with exterior differentiation form such a complex. We put $\Omega^k(M) = 0$ for $k < 0$.

Definition 13.2. The *de Rham cohomology* in degree k is the quotient

$$H^k(M) = \frac{\ker(\mathrm{d}\colon \Omega^k(M) \to \Omega^{k+1}(M))}{\operatorname{im}(\mathrm{d}\colon \Omega^{k-1}(M) \to \Omega^k(M))}.$$

Definition 13.3. The *cohomology class* $[\alpha]$ of a closed form consists of all closed forms differing from α by an exact form.

In positive degree, $[\alpha] = 0$ means precisely that α has a global primitive. Lecture 6 formed homology as the kernel of a boundary map modulo the image of the preceding boundary map. Here the arrows reverse direction: exterior differentiation raises degree, while the chain boundary lowers it. The identity $\mathrm{d}^2 = 0$ replaces $\partial^2 = 0$. Closed forms play the

role of cycles, and exact forms play the role of boundaries. The superscript distinguishes cohomology from homology.

In degree zero there are no forms of degree -1, so the only exact form is zero. A function is closed exactly when all its partial derivatives vanish. It is then constant on connected coordinate balls, hence locally constant. Conversely a locally constant function has zero derivative. In particular, $H^0(M) = \mathbb{R}$ when M is connected and nonempty. For a disconnected manifold, each component contributes an independently chosen constant. With infinitely many components this is a product of copies of $\mathbb{R}$.

For example, on the punctured plane the angular form is

$$\alpha_{\text{ang}} = \frac{-y\,\mathrm{d}x + x\,\mathrm{d}y}{x^2 + y^2}.$$

Lecture 10 proved that this form is closed and has integral 2π around the counter-clockwise unit circle. An exact form has zero integral around that circle by Stokes. Hence $[\alpha_{\text{ang}}] \neq 0$.

Definition 13.4. A *period* of a closed k-form is its integral over a closed oriented k-manifold mapped smoothly into M, using pullback.

Such a mapped manifold is a geometric cycle. Stokes makes every period of an exact form zero, so periods depend only on the class. These geometric cycles suffice here. Lecture 14 relates forms to the singular cycles of Lecture 6.

Definition 13.5. The *induced pullback* of a smooth map $f\colon M \to N$ is given by

$$f^*\colon H^k(N) \longrightarrow H^k(M), \qquad f^*[\alpha] = [f^*\alpha].$$

Pullback commutes with d, so it preserves closed forms and exact differences. This proves that the induced map is well defined. Pullback reverses composition: $(g \circ f)^* = f^* g^*$, and the identity map induces the identity. The wedge product also descends to classes. For closed β, $(\mathrm{d}\eta) \wedge \beta = \mathrm{d}(\eta \wedge \beta)$. For closed α of degree k, $\alpha \wedge \mathrm{d}\xi = (-1)^k \mathrm{d}(\alpha \wedge \xi)$. Changing either representative therefore changes the product only by an exact form. We write $H^*(M)$ for all the degrees together. For example, on $T^2 = S^1 \times S^1$ let ν_1 and ν_2 be the pullbacks of the period-one angular form from the two factors. In angular coordinates these are $\mathrm{d}\theta/(2\pi)$ and $\mathrm{d}\phi/(2\pi)$. With the product orientation their wedge has period

$$\int_{T^2} \nu_1 \wedge \nu_2 = \frac{1}{(2\pi)^2} \int_0^{2\pi} \int_0^{2\pi} \mathrm{d}\theta\, \mathrm{d}\phi = 1.$$

Thus $[\nu_1] \wedge [\nu_2]$ is a nonzero degree-two class.

13.2 The homotopy operator and the Poincaré lemma

Definition 13.6. An open set $U \subset \mathbb{R}^n$ is *star-shaped about* 0 if $0 \in U$ and $tx \in U$ for every $x \in U$ and $t \in [0, 1]$.

For a closed 1-form α on an open set star-shaped about the origin, integration along the segment from 0 to x gives a potential function:

$$f(x) = \int_0^1 \alpha_{tx}(x)\, \mathrm{d}t.$$

The homotopy operator extends this construction to forms of every degree.

Definition 13.7. A *smooth homotopy* between smooth maps F_0 and F_1 from M to N is a smooth map $F\colon [0,1]\times M \to N$ with $F_t(p) = F(t,p)$ and those endpoint maps. Smoothness at the endpoints means local smooth extension, as in Lecture 12.

No inverse for any F_t is required. For a k-form α on N, the pulled-back form decomposes uniquely as

$$F^*\alpha = \mathrm{d}t \wedge a_t + b_t,$$

where a_t and b_t are forms on M of degrees $k-1$ and k. Here a_t is the restriction of $\iota_{\partial_t} F^*\alpha$ to the slice at time t, and $b_t = F_t^*\alpha$.

Definition 13.8. The *homotopy operator* is given by integrating the time-containing coefficients:

$$K_F\colon \Omega^k(N) \longrightarrow \Omega^{k-1}(M), \qquad K_F\alpha = \int_0^1 a_t\,\mathrm{d}t \quad (k \geq 1),$$
$$K_F f = 0 \quad (k = 0).$$

The integral is smooth. Near each point, compactness of the time interval permits differentiating the smooth coefficients under the integral. The intrinsic definition by contraction makes the chart definitions agree.

Proposition 13.9 (Homotopy operator identity)**.** *For every smooth homotopy F and every differential form α,*

$$F_1^*\alpha - F_0^*\alpha = \mathrm{d}K_F\alpha + K_F\mathrm{d}\alpha.$$

Proof. Write d_M for differentiation in M. The product rule gives

$$\mathrm{d}(F^*\alpha) = \mathrm{d}t \wedge (\partial_t b_t - \mathrm{d}_M a_t) + \mathrm{d}_M b_t.$$

The minus sign comes from moving d_M past $\mathrm{d}t$. Since $\mathrm{d}F^*\alpha = F^*\mathrm{d}\alpha$, the time-containing coefficient on the right is the one integrated in $K_F\mathrm{d}\alpha$. Thus

$$K_F\mathrm{d}\alpha = \int_0^1 \partial_t b_t\,\mathrm{d}t - \mathrm{d}_M \int_0^1 a_t\,\mathrm{d}t = b_1 - b_0 - \mathrm{d}_M K_F\alpha.$$

The fundamental theorem of calculus gives the endpoint difference. This is the claimed identity. For functions, the same computation has $a_t = 0$ and is the ordinary endpoint formula. ∎

For a flow $F_t = \varphi_t$ generated by X, the time coefficient is $a_t = \varphi_t^*(\iota_X\alpha)$. Indeed, $\partial_t F(t,p) = X_{F(t,p)}$ fills the first argument of α. The homotopy identity is then the integral of Lecture 11's Cartan formula $L_X = \mathrm{d}\iota_X + \iota_X\mathrm{d}$. For a smooth oriented path σ in M, Stokes on its swept square gives

$$\int_\sigma (F_1^*\alpha - F_0^*\alpha) = \int_{\partial\sigma} K_F\alpha + \int_{[0,1]\times\sigma} F^*\mathrm{d}\alpha \qquad (\deg\alpha = 1).$$

The square has time-first orientation. Its side edges account for the endpoint values of $K_F\alpha$, as the figure shows.

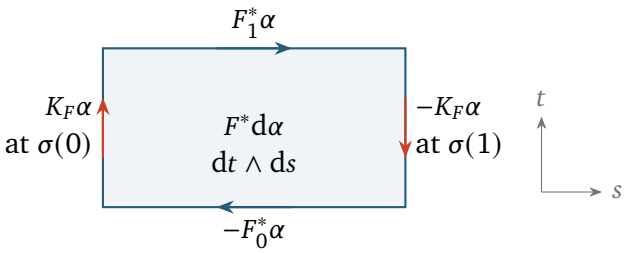


Figure 13.1: The homotopy sweeps a path into a square oriented by $\mathrm{d}t \wedge \mathrm{d}s$. The top and bottom give the endpoint pullbacks, and the sides give the boundary values of K_F.

Definition 13.10. Two maps of complexes whose difference is $\mathrm{d}K + K\mathrm{d}$ are called *cochain homotopic*.

The displayed identity constructs that relation for the two endpoint pullbacks.

For a star-shaped open set U, the radial homotopy is given by $F(t, x) = tx$. The figure shows its segments from 0 to x.

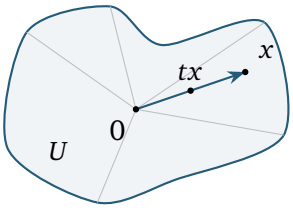


Figure 13.2: In a star-shaped set, each segment from 0 to x stays inside U. The contraction is given by $F(t, x) = tx$.

For this homotopy the operator has the explicit formula, for $k \geq 1$,

$$(K\alpha)_x(v_1, \ldots, v_{k-1}) = \int_0^1 t^{k-1}\alpha_{tx}(x, v_1, \ldots, v_{k-1})\,\mathrm{d}t. \tag{13.1}$$

Indeed, the time derivative of F is x, while each spatial tangent vector is multiplied by t. This gives the power t^{k-1}. Near a fixed $x \in U$, compactness of its segment and openness of U give $\varepsilon > 0$ and a neighbourhood of x such that $tx' \in U$ for every point x' in that neighbourhood and $t \in (-\varepsilon, 1 + \varepsilon)$. This gives the required local smooth extension in time. Differentiating the coefficients under the integral therefore defines a smooth form on U.

Theorem 13.11 (Poincaré lemma). *On a star-shaped open set, every closed k-form with $k \geq 1$ is exact. Such a set has degree-zero cohomology consisting of the constant functions.*

Proof. After translating the star centre, use $F(t, x) = tx$. The map F_1 is the identity. The map F_0 is constant, so its differential is zero and its pullback on positive-degree forms is zero. For closed α the operator identity consequently gives $\alpha = \mathrm{d}K\alpha$. In degree zero it instead gives $f - f(0) = K\mathrm{d}f$. A closed function is therefore constant. ∎

For example, $\alpha = 2x\mathrm{d}x + 2y\mathrm{d}y$ on $\mathbb{R}^2$ has $K\alpha = \int_0^1 2t(x^2+y^2)\,\mathrm{d}t = x^2+y^2$. For a degree-two example on $\mathbb{R}^2$, take $\alpha = \mathrm{d}x \wedge \mathrm{d}y$. For a tangent vector $v = (v_x, v_y)$, the radial formula gives

$$(K\alpha)_{(x,y)}(v) = \int_0^1 t(xv_y - yv_x)\,\mathrm{d}t = \frac{xv_y - yv_x}{2}.$$

The radial vector fills the first argument. The remaining tangent vector contributes the factor t. Hence

$$K(\mathrm{d}x \wedge \mathrm{d}y) = \frac{x\mathrm{d}y - y\mathrm{d}x}{2}, \qquad \mathrm{d}K(\mathrm{d}x \wedge \mathrm{d}y) = \frac{\mathrm{d}x \wedge \mathrm{d}y - \mathrm{d}y \wedge \mathrm{d}x}{2} = \mathrm{d}x \wedge \mathrm{d}y.$$

In $\mathbb{R}^3$, write a closed 2-form as $B_x\mathrm{d}y \wedge \mathrm{d}z + B_y\mathrm{d}z \wedge \mathrm{d}x + B_z\mathrm{d}x \wedge \mathrm{d}y$. Its closedness means $\nabla \cdot \mathbf{B} = 0$, by Lecture 10. On a region star-shaped about 0, its radial primitive is the 1-form $\mathbf{A} \cdot \mathrm{d}\mathbf{x}$, where

$$\mathbf{A}(x) = \int_0^1 t\,\mathbf{B}(tx) \times x\,\mathrm{d}t.$$

This follows from the determinant identity $\mathbf{B} \cdot (x \times v) = (\mathbf{B} \times x) \cdot v$. For a constant field, the integral gives $\mathbf{A}(x) = \mathbf{B} \times x/2$.

The punctured plane is not star-shaped. Its radial paths end at a missing point, so the homotopy does not have its claimed target there. A formal insertion into α_{ang} would even give zero, because $-yx + xy = 0$. It cannot supply a primitive of that nonzero form.

13.3 Homotopy invariance

Theorem 13.12 (Homotopy invariance · *proved given cited smoothing; smooth homotopies proved directly*)**.** *Smoothly homotopic maps induce the same maps on de Rham cohomology. The same holds for smooth maps that are continuously homotopic. Smooth manifolds with the same continuous homotopy type have isomorphic cohomology in every degree.*

Proof. For a smooth homotopy and a closed form α, the operator identity gives $F_1^*\alpha - F_0^*\alpha = \mathrm{d}K_F\alpha$. The two pullbacks therefore represent the same class.

For the continuous case we use the following smooth approximation results. Every continuous map between smooth manifolds is homotopic to a smooth map. Two homotopic smooth maps admit a smooth homotopy. These are cited from Lee, *Introduction to Smooth Manifolds*, second edition, Theorems 6.26 and 6.29, pp. 141–143. Choose a smooth representative of a continuous map's homotopy class and use its pullback. Two choices induce the same map by the second cited result and the smooth proof above. Composition and identities are preserved because compositions of smooth representatives represent the continuous compositions. If continuous maps $f\colon M \to N$ and $g\colon N \to M$ are homotopy inverses, their composites represent identity maps. Contravariance then gives $f^*g^* = \mathrm{id}$ and $g^*f^* = \mathrm{id}$ on the respective cohomology spaces. Thus these maps are inverse isomorphisms. ∎

A contractible manifold consequently has the cohomology of a point: $\mathbb{R}$ in degree zero and zero in positive degrees. For $n \geq 1$, the punctured space $\mathbb{R}^n \setminus \{0\}$ retracts onto S^{n-1} by $r(x) = x/\|x\|$. With i the inclusion, $r \circ i = \mathrm{id}$. A smooth homotopy from the identity to $i \circ r$ is

$$F(t, x) = \left((1-t)^2 + \frac{t^2}{\|x\|}\right) x.$$

We use squares in this coefficient to obtain a homotopy defined for all real t, including smooth extensions past both endpoints. The coefficient is positive for every real t, since its two nonnegative summands never vanish together. Hence the homotopy stays in the punctured space. It has the required endpoints. This proves $H^*(\mathbb{R}^n \setminus \{0\}) \cong H^*(S^{n-1})$.

Self-check. State what a de Rham cohomology class is and why we introduced it. Explain how a nonzero period distinguishes a class from zero.

Answer. A class is a closed form modulo addition of exact forms. In positive degree, the zero class consists of forms with a global primitive. We introduced the quotient to measure the obstruction to such a primitive. Stokes makes every period of an exact form zero, so a nonzero period proves that a closed form represents a nonzero class.

Unit 13B Gluing forms and computing sphere cohomology

13.4 Exact sequences and Mayer–Vietoris

Lecture 5 used van Kampen to compute the fundamental group of a union from two open sets and their overlap. Mayer–Vietoris uses the same cover to compute cohomology. We restrict forms to the pieces, compare them on the overlap, and differentiate local lifts to recover a global class.

Definition 13.13. A sequence of vector spaces and linear maps is *exact* at a term if the image of the incoming map equals the kernel of the outgoing map.

Definition 13.14. A *short exact sequence* has the form $0 \to A \to B \to C \to 0$.

Its first nonzero map is injective and its last is surjective. For a sequence, exactness is a property of adjacent maps. For a form, exactness means that the form is an exterior derivative. The uses are related: a cochain complex is exact at a degree precisely when every closed element in that degree is exact.

Definition 13.15. The *cokernel* of a linear map $T\colon A \to B$ is the quotient

$$\operatorname{coker} T = B/\operatorname{im} T.$$

Let $M = U \cup V$ be an open cover and set $W = U \cap V$. Restriction gives a short exact sequence of form spaces in each degree:

$$0 \longrightarrow \Omega^k(M) \xrightarrow{i} \Omega^k(U) \oplus \Omega^k(V) \xrightarrow{j} \Omega^k(W) \longrightarrow 0, \qquad j(a,b) = b|_W - a|_W.$$

The map i restricts a global form to both sets. It is injective because these sets cover M. A pair is in $\ker j$ exactly when its forms agree on the overlap, which is precisely when they glue to a global form. To prove surjectivity, choose a partition $\rho_U + \rho_V = 1$ subordinate to the cover, supplied by Theorem 12.8. For $\zeta \in \Omega^k(W)$ take

$$a = -\operatorname{ext}_U(\rho_V \zeta), \qquad b = \operatorname{ext}_V(\rho_U \zeta).$$

Here ext means extension by zero from the overlap. This extension is smooth: near a point of $U \setminus V$, the support condition makes ρ_V zero, and similarly on $V \setminus U$. Their difference on W is $(\rho_U + \rho_V)\zeta = \zeta$. All restriction and difference maps commute with differentiation.

For a closed overlap form ζ, choose any lifts with $j(a,b) = \zeta$. Then $\mathrm{d}b - \mathrm{d}a = \mathrm{d}\zeta = 0$ on W. The derivatives glue to a global closed form c of degree $k+1$.

Definition 13.16. The *connecting homomorphism* is given by

$$\delta_{\mathrm{MV}}\colon H^k(W) \longrightarrow H^{k+1}(M), \qquad \delta_{\mathrm{MV}}[\zeta] = [c], \quad c|_U = \mathrm{d}a, \quad c|_V = \mathrm{d}b.$$

There is no additional sign. The subscript identifies this as the Mayer–Vietoris map. Different lifts differ by restrictions of a global form η, so their resulting c differs by $\mathrm{d}\eta$. If ζ changes by $\mathrm{d}\xi$, lift ξ to (a', b') and replace (a, b) by $(a + \mathrm{d}a', b + \mathrm{d}b')$. The resulting c is unchanged. In degree zero there is no such exact change. These observations prove well-definedness. Adding or scaling lifts proves linearity.

Theorem 13.17 (Mayer–Vietoris · *exactness sketched; connecting map and overlap exactness proved given partitions of unity*)**.** *The induced maps and connecting homomorphisms form an exact sequence*

$$\begin{aligned} 0 \longrightarrow H^0(M) \xrightarrow{i^*} H^0(U) \oplus H^0(V) \xrightarrow{j^*} H^0(W) \xrightarrow{\delta_{\mathrm{MV}}} H^1(M) \longrightarrow \cdots, \\ \cdots \longrightarrow H^k(M) \xrightarrow{i^*} H^k(U) \oplus H^k(V) \xrightarrow{j^*} H^k(W) \xrightarrow{\delta_{\mathrm{MV}}} H^{k+1}(M) \longrightarrow \cdots. \end{aligned} \tag{13.2}$$

Proof at the overlap term and sketch elsewhere. If $\zeta = b - a$ for closed lifts, its connecting form is zero. Hence $\operatorname{im} j^* \subset \ker \delta_{\mathrm{MV}}$. Conversely, if $\delta_{\mathrm{MV}}[\zeta] = 0$, its glued form is $c = \mathrm{d}\eta$ for a global η. Subtract the restrictions of η from both lifts. Their derivatives now vanish and their difference remains ζ. Thus $[\zeta] \in \operatorname{im} j^*$, proving exactness at $H^k(W)$.

At the pair term, an exact difference of closed representatives can be removed by lifting its primitive through j. The adjusted representatives agree and glue. At the global term, local primitives of a class restricting to zero have a closed difference on W whose connecting class is the original class. The subsection on the remaining exactness arguments supplies the representative and degree-zero details. This is the cochain construction of Tu, §25–§26, especially (26.1), pp. 288–291. ∎

The remaining exactness arguments *(optional)*

Later sphere and bundle computations do not depend on the proof details in this subsection.

Exactness at the pair and global terms. At the pair term, restrictions of a global closed form have zero difference. Thus $\operatorname{im} i^* \subset \ker j^*$. Conversely, let (a, b) be closed representatives of a class in $\ker j^*$. For $k \geq 1$, their difference is $b - a = \mathrm{d}\xi$ on W. Surjectivity of j supplies forms r and s of degree $k - 1$ with $s - r = \xi$ on W. The adjusted forms satisfy

$$(b - \mathrm{d}s) - (a - \mathrm{d}r) = \mathrm{d}\xi - \mathrm{d}(s - r) = 0.$$

They are closed and represent the original classes on U and V. They therefore glue to a global closed form. This proves the reverse inclusion. For $k = 0$, a zero difference class means $b - a = 0$, since no nonzero 0-form is exact. The functions already agree and glue.

At the global term of degree $k \geq 1$, a connecting form restricts to $\mathrm{d}a$ and $\mathrm{d}b$. Its restrictions are exact, so $\operatorname{im} \delta_{\mathrm{MV}} \subset \ker i^*$. Conversely, let a global closed k-form c have zero restriction classes. Choose local primitives a and b with $c|_U = \mathrm{d}a$ and $c|_V = \mathrm{d}b$. Their difference $\zeta = b - a$ is closed on W, because $\mathrm{d}\zeta = c|_W - c|_W = 0$. These primitives are lifts of ζ, so $\delta_{\mathrm{MV}}[\zeta] = [c]$. This proves the reverse inclusion. In degree zero the incoming map is zero. A closed function with zero restriction classes vanishes on each cover member, hence vanishes on M. Thus i^* is injective also in degree zero. ∎

A circle connecting class

We use the ordered cover $U = S^1 \setminus \{S\}$ and $V = S^1 \setminus \{N\}$, where $S = (0, -1)$ and $N = (0, 1)$. The overlap consists of the left arc W_- and the right arc W_+. The counter-clockwise orientation

traverses W_+ from S to N. Write z for the vertical coordinate in this circle calculation. An explicit smooth partition is $\rho_U = h(z)$ and $\rho_V = 1 - h(z)$, where

$$E(s) = \begin{cases} e^{-1/s}, & s > 0, \\ 0, & s \le 0, \end{cases} \qquad h(z) = \frac{E(z + 1/2)}{E(z + 1/2) + E(1/2 - z)}.$$

For $s > 0$, each derivative of E is a polynomial in $1/s$ times $e^{-1/s}$. Each expression tends to zero as s decreases to zero. Thus these derivatives match the zero derivatives on the negative half-line, so E is smooth. The denominator defining h is positive for every $z \in \mathbb{R}$. The support of ρ_U avoids S, and the support of ρ_V avoids N. Take the overlap function ζ equal to 0 on W_- and 1 on W_+. Write c for its glued connecting form. The figure plots this partition and the coefficient of c on W_+.

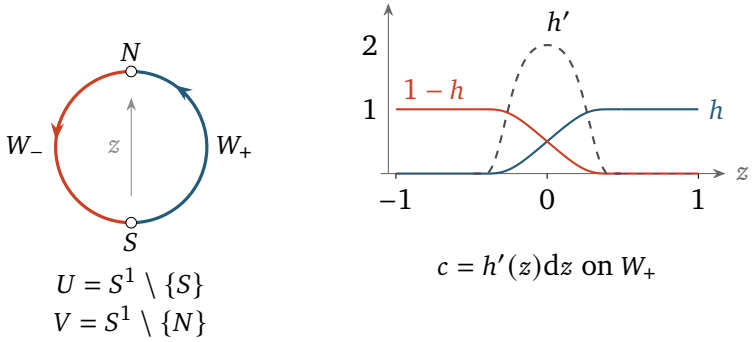


Figure 13.3: On the right overlap arc, increasing height follows the positive circle orientation. The connecting form has coefficient $h'(z)$ there and is zero on the left arc.

Example 13.18 (An overlap function with period-one image)**.** The overlap function ζ is closed, since it is locally constant. On W_+ choose $a = -\rho_V$ and $b = \rho_U$. On W_- choose both lifts to be zero. Extend a across N by zero, and extend b across S by zero. The support conditions make these extensions smooth. Their difference is ζ. On W_+ their derivatives agree:

$$\mathrm{d}a = -\mathrm{d}\rho_V = \mathrm{d}\rho_U = \mathrm{d}b = h'(z)\mathrm{d}z.$$

The glued form c vanishes on W_- and near both poles. Its period is

$$\int_{S^1} c = \int_{-1}^{1} h'(z)\,\mathrm{d}z = h(1) - h(-1) = 1.$$

Thus $\delta_{\mathrm{MV}}[\zeta] = [c]$ is a nonzero circle class.

13.5 Cohomology of spheres

Remove the south pole to obtain $U \subset S^n$ and the north pole to obtain V, for $n \ge 1$. Each set is diffeomorphic to $\mathbb{R}^n$ by stereographic projection. Their overlap has the product description in the figure.

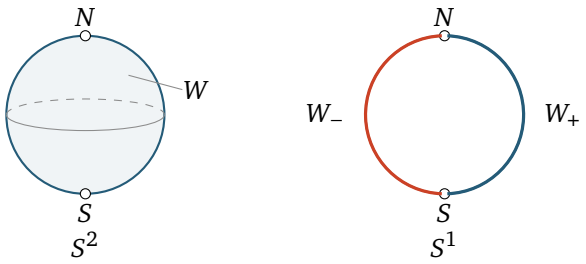


Figure 13.4: Each open set omits one pole. Their overlap is the whole sphere except the two marked poles, an open cylinder for S^2. For S^1 it has two components.

Write a sphere point as $(x', z) \in \mathbb{R}^n \times \mathbb{R}$. On W, the map $(x', z) \mapsto (x'/\sqrt{1-z^2}, z)$ is a diffeomorphism onto $S^{n-1} \times (-1, 1)$, with inverse $(u, z) \mapsto (\sqrt{1-z^2}u, z)$. Contracting the second coordinate to zero retracts W onto S^{n-1}. The two contractible cover members have zero positive-degree cohomology.

Theorem 13.19 (Sphere computation · *proved given Mayer–Vietoris exactness*)**.** *For $n \geq 1$ and $k \geq 0$,*

$$H^k(S^n) \cong \begin{cases} \mathbb{R}, & k = 0 \text{ or } k = n, \\ 0, & \text{otherwise.} \end{cases}$$

For S^0, degree zero is $\mathbb{R}^2$ and all higher degrees vanish. For $n \geq 1$, punctured $\mathbb{R}^n$ has the cohomology of S^{n-1}.

Proof. The space S^0 is two points. A function assigns one value to each point, and all positive-degree forms vanish. This proves the initial case. For $n = 1$, the overlap has two contractible components. The beginning of Mayer–Vietoris is

$$0 \longrightarrow H^0(S^1) \longrightarrow \mathbb{R}^2 \xrightarrow{(a,b)\mapsto(b-a,b-a)} \mathbb{R}^2 \xrightarrow{\delta_{\mathrm{MV}}} H^1(S^1) \longrightarrow 0.$$

The map after δ_{MV} lands in zero, so exactness makes δ_{MV} surjective. Exactness at its source gives

$$\ker \delta_{\mathrm{MV}} = \operatorname{im} j^* = \{(c, c) : c \in \mathbb{R}\}.$$

Hence the induced quotient map is an isomorphism

$$H^1(S^1) \cong \mathbb{R}^2/\{(c, c) : c \in \mathbb{R}\}.$$

This quotient is the cokernel of $j^* : H^0(U) \oplus H^0(V) \to H^0(W)$. In the order (W_-, W_+), the value $v - u$ labels the class of (u, v) modulo the diagonal. Its kernel is precisely that diagonal, so the quotient is one-dimensional. The kernel of j^* is also the diagonal, which gives $H^0(S^1) = \mathbb{R}$. The figure shows the quotient coordinate.

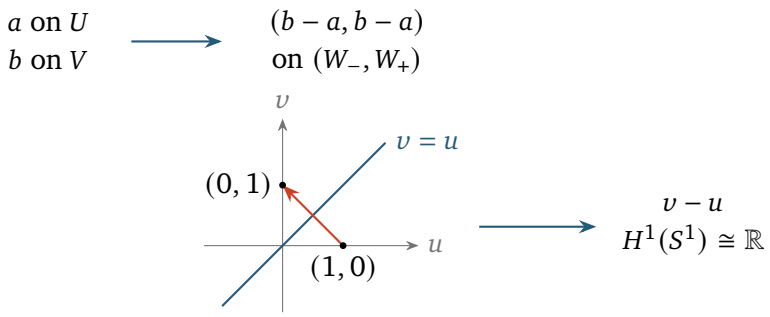


Figure 13.5: Constant functions on U and V give the same difference on both overlap components. The coordinate $v - u$ is unchanged by adding a diagonal vector, so it identifies the quotient with $\mathbb{R}$.

Higher degrees vanish because the circle has dimension one.

For $n \geq 2$, the overlap is connected, so its degree-zero group is $\mathbb{R}$. The difference map $\mathbb{R}^2 \to \mathbb{R}$ is surjective with diagonal kernel. Exactness gives $H^0(S^n) = \mathbb{R}$ and $H^1(S^n) = 0$. For $k \geq 2$, the zero positive-degree groups of U and V give

$$0 \longrightarrow H^{k-1}(W) \xrightarrow{\delta_{\mathrm{MV}}} H^k(S^n) \longrightarrow 0.$$

The left zero forces injectivity, and the right zero forces surjectivity. Thus δ_{MV} is an isomorphism and $H^k(S^n) \cong H^{k-1}(S^{n-1})$. Induction gives precisely one nonzero positive-degree group, in degree n. The explicit retraction in §13.3 proves the punctured-space assertion. In particular, punctured $\mathbb{R}$ has two-dimensional degree-zero cohomology, whereas punctured $\mathbb{R}^n$ for $n \geq 2$ is connected. ∎

13.6 The area class and local primitives on the sphere

On the outward-oriented unit sphere let β be Lecture 10's area form, $\sin\theta\, \mathrm{d}\theta \wedge \mathrm{d}\phi$. It is also the form denoted by μ for $n = 2$ in Lecture 12. Write β_C for Lecture 12's Coulomb flux form on $\mathbb{R}^3 \setminus \{0\}$. Its restriction to the unit sphere is β. We compute its class using the ordered cover above: $U = S^2 \setminus \{\text{south}\}$ and $V = S^2 \setminus \{\text{north}\}$. Write the local 1-forms as a_U and a_V, and their common derivative as c. Their difference and derivatives are related as in the diagram.

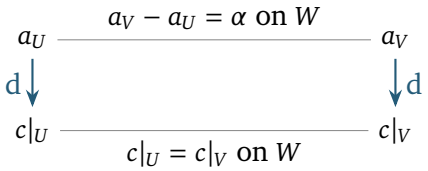


Figure 13.6: Local 1-forms differ by a closed form on W. Their derivatives agree and glue to the 2-form representing the connecting class.

Example 13.20 (The normalized sphere class)**.** The sphere computation gives $H^2(S^2) = \mathbb{R}$ and $H^1(S^2) = 0$. The form $\beta/(4\pi)$ is closed, since every 3-form on a surface vanishes. Its period is

$$\int_{S^2} \frac{\beta}{4\pi} = \frac{1}{4\pi} \int_0^{2\pi} \int_0^{\pi} \sin\theta\, \mathrm{d}\theta\, \mathrm{d}\phi = 1.$$

It is therefore not exact and its class spans $H^2(S^2)$.

The nonzero period prevents a global primitive of $\beta/(4\pi)$. Local primitives still exist. Their overlap difference computes the sign of the connecting map and supplies the cap potentials used in the magnetic monopole model. On W, the globally defined angular 1-form is $\alpha = \mathrm{d}\phi/(2\pi)$, where $\mathrm{d}\phi$ denotes the angular form, not the differential of a globally defined real angle. Choose lifts

$$a_U = -\frac{x\mathrm{d}y - y\mathrm{d}x}{4\pi(1+z)}, \qquad a_V = \frac{x\mathrm{d}y - y\mathrm{d}x}{4\pi(1-z)}.$$

Their denominators do not vanish on their respective domains, so these forms are smooth also at the opposite poles. On the overlap, $x\mathrm{d}y - y\mathrm{d}x = (1-z^2)\mathrm{d}\phi$. Consequently $a_U = -(1-z)\mathrm{d}\phi/(4\pi)$ and $a_V = (1+z)\mathrm{d}\phi/(4\pi)$, giving $a_V - a_U = \alpha$. Their common derivative is

$$\mathrm{d}a_U = \mathrm{d}a_V = \frac{\mathrm{d}z \wedge \mathrm{d}\phi}{4\pi} = -\frac{\sin\theta\,\mathrm{d}\theta \wedge \mathrm{d}\phi}{4\pi} = -\frac{\beta}{4\pi}.$$

Equality extends over each pole by smoothness. Thus $\delta_{\mathrm{MV}}[\mathrm{d}\phi/(2\pi)] = -[\beta/(4\pi)]$, with the order U, V fixed in §13.5. The positive period-one class is $[\beta/(4\pi)]$. A nonzero period obstructs joining its local primitives to a global one. The closed Coulomb form β_C restricts to the area form β. Restriction is an isomorphism on cohomology, inverse to the radial retraction pullback. It sends $[\beta_C/(4\pi)]$ to $[\beta/(4\pi)]$. Hence $[\beta_C/(4\pi)]$ spans $H^2(\mathbb{R}^3 \setminus \{0\})$, with sphere period one.

Cap potentials and the magnetic monopole

Physical input. A magnetic field is represented by a real flux 2-form F_{em}. A local real magnetic potential A_{em} satisfies $F_{\mathrm{em}} = \mathrm{d}A_{\mathrm{em}}$. We model a point magnetic source at the origin by $F_{\mathrm{em}} = (\Phi_{\mathrm{m}}/(4\pi))\beta_C$ on $\mathbb{R}^3 \setminus \{0\}$. The parameter Φ_{m} is its outward sphere flux.

The cap primitives above give real potentials on the unit sphere:

$$A^{(N)}_{\mathrm{em}} = -\Phi_{\mathrm{m}} a_U, \qquad A^{(S)}_{\mathrm{em}} = -\Phi_{\mathrm{m}} a_V, \qquad \mathrm{d}A^{(N)}_{\mathrm{em}} = \mathrm{d}A^{(S)}_{\mathrm{em}} = \frac{\Phi_{\mathrm{m}}}{4\pi}\beta.$$

The north chart is U, and the south chart is V. For nonzero flux, Stokes excludes a global potential on the sphere. The overlap difference $a_V - a_U = \mathrm{d}\phi/(2\pi)$ is closed. It is a differential on each local angle patch, but its period one prevents a single global real angle on the overlap.

For comparison with Lecture 22, let $q \neq 0$ be a particle's signed charge and set $n = q\Phi_{\mathrm{m}}/(2\pi\hbar)$, initially a real parameter. That lecture uses imaginary local potentials, obtained by the rescaling

$$A_N = 2\pi i n\, a_U = -\frac{iq}{\hbar} A^{(N)}_{\mathrm{em}}, \qquad A_S = 2\pi i n\, a_V = -\frac{iq}{\hbar} A^{(S)}_{\mathrm{em}}.$$

Their common derivative is $-in\beta/2$, and their difference is $A_S - A_N = in\mathrm{d}\phi$. Lecture 22 derives the integer condition on n from compatibility of local charged states. Real de Rham cohomology alone allows every real multiple of $[\beta/(4\pi)]$.

Radially extending the northern potential excludes the negative z-axis. This excluded half-axis is called the *Dirac string* in the singular potential description. For $\Phi_{\mathrm{m}} > 0$, a small increasing-azimuth circle near this axis has limiting circulation Φ_{m}. In the thin-solenoid representation this is flux toward the origin along the negative axis. The string represents

the singular choice of potential, not an additional field in the two-chart model. The figure distinguishes the signed circulations on equatorial test circles from this string.

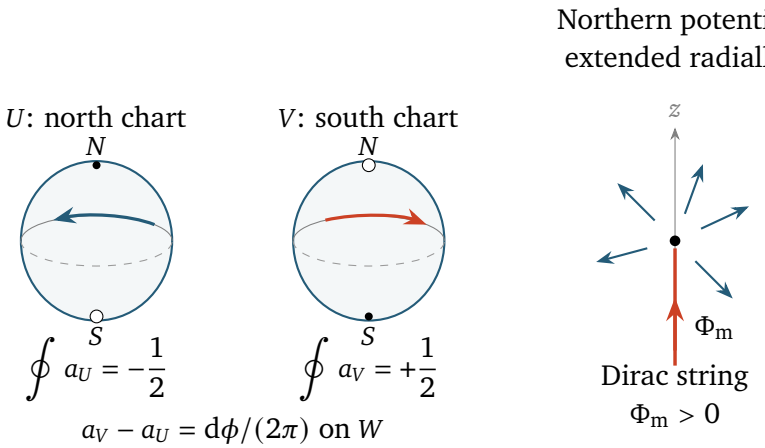


Figure 13.7: The cap arrows show the signs of the real 1-forms on equatorial test circles: a_U has negative azimuthal circulation and a_V positive circulation. For positive flux the singular northern-potential description has a string on the negative z-axis directed toward the origin.

Real coefficients lose torsion information. A torsion element is one annihilated by a positive integer. No real vector space has a nonzero torsion element. In particular, Lecture 4's order-two loop in SO(3) cannot be detected by a closed real 1-form's period. Use smooth loops constant near their endpoints, so their concatenation is smooth. Periods of a closed 1-form are invariant under smooth loop homotopy by Stokes on the cylinder. Continuous homotopies can be smoothed by the cited theorem. Periods add under loop composition, so twice that loop's period is zero and its real period is zero.

Self-check. For the ordered circle cover above, let ζ have values $-5/2$ on W_- and $7/3$ on W_+. Compute the period of $\delta_{\mathrm{MV}}[\zeta]$. Decide whether the overlap values $(-1/2, 13/3)$ give the same class.

Answer. For overlap values (u, v), subtracting (u, u) leaves $(0, v-u)$. The period-one calculation therefore makes the period equal to $7/3 - (-5/2) = 29/6$. The second pair differs from the first by $(2, 2)$, which is in $\operatorname{im} j^*$. The connecting classes are equal.

Self-check. Find $H^1(\mathbb{R}^2 \setminus \{0\})$ and a generator of period one on the counter-clockwise unit circle.

Answer. Radial retraction gives the cohomology of S^1, so this space is one-dimensional. The angular form is closed by Lecture 10. A nonzero period rules out an exact form by Stokes. The class $[\alpha_{\mathrm{ang}}/(2\pi)]$ has period one by Lecture 10. It is nonzero and therefore spans the space.

Summary. De Rham cohomology consists of closed forms modulo exact forms, over $\mathbb{R}$. The homotopy operator proves local exactness in positive degree on star-shaped sets and invariance under smooth homotopy. Cited smoothing extends invariance to continuous homotopy type. Mayer–Vietoris computes sphere cohomology from two contractible sets. On S^2, the area class $[\beta/(4\pi)]$ has period one. Lecture 14 relates periods to degree and singular homology.

Reading. Tu, *An Introduction to Manifolds*, §24–§29, especially §26, pp. 288–291, and §29, pp. 311–317. Lee, *Introduction to Smooth Manifolds*, second edition, Theorems 6.26 and 6.29, pp. 141–143.